

CALIBER 2026

Future-Ready Operational Excellence: Knowledge, Reliability, and Smart Manufacturing

For eligibility, registration, key dates, and contact information, please refer to the official CALIBER 2026 Booklet. This casebook covers case content and submission only.

Company Background

PT Chandra Asri Pacific Tbk (Chandra Asri Group) is an energy, chemical, and infrastructure solutions company in Southeast Asia. Operating complex and asset-intensive petrochemical facilities, the Company relies on technical knowledge, equipment reliability, supply-chain readiness, and efficient resource utilization to maintain safe and continuous operations.

As its plants, systems, and operational data continue to grow, the Company is strengthening its digital transformation agenda by integrating data, analytics, and artificial intelligence into manufacturing processes. This transformation is intended to support faster decision-making, improve operational visibility, preserve critical plant knowledge, and enhance collaboration across Production, Maintenance, Reliability, Warehouse, Procurement, Energy, and Health, Safety, Environment (HSE) functions.



Executive Summary

Future-Ready Operational Excellence: Knowledge, Reliability, and Smart Manufacturing grand theme focuses on applying artificial intelligence to address critical manufacturing challenges within a petrochemical environment.

Participants are expected to propose practical and scalable AI solutions that transform fragmented operational data into actionable insight. The proposed solutions may support technical knowledge retention, predictive maintenance, spare-part inventory optimization, energy and emissions monitoring, production forecasting, anomaly detection, and cross-functional decision-making.

The solutions should integrate relevant plant and enterprise data, provide transparent and traceable recommendations, and consider human oversight, data security, system reliability, and measurable business impact. Through this grand theme, the Company aims to accelerate the development of safer, more reliable, efficient, and intelligent manufacturing operations.



Please choose only one of the two use cases provided in the following section and build your solution based on it.



CASE 1

Manufacturing Knowledge Hub (AI-Powered Knowledge Integration)

Topic

This case focuses on reducing improper execution risk caused by scattered knowledge source, missing tacit knowledge, difficulty in finding trusted information, and disconnected operational data.

Background

In a chemical manufacturing plant, engineers need to make quick and accurate decisions during start-up, troubleshooting, maintenance activities, and abnormal operating conditions. However, the information they need is often spread across different sources: SOPs are stored in one system, datasheets in another, P&IDs in document folders, maintenance logs in separate records, and real-time operating data in plant systems.

Because these knowledge sources are not connected, engineers may spend too much time searching for trusted information or asking experienced personnel for confirmation. In some cases, missing tacit knowledge, outdated documents, or incomplete context may cause teams to misinterpret procedures or fail to follow the correct execution steps.

As a result, plant operations may face longer troubleshooting time, repeated engineering rework, higher safety risk, delayed start-up, and inconsistent decision-making across Process, Maintenance, Engineering, Operation, and HSE functions.

Problem Statement

Participants are challenged to design an AI-enabled Manufacturing Knowledge Hub that scattered plant knowledge from SOPs, datasheets, P&IDs, EDMS, AIMS, maintenance logs, incident reports, and real-time operational data into one trusted, structured, and traceable knowledge ecosystem. The solution should help engineers find verified information faster, reduce improper execution, support troubleshooting, and improve reliability, safety, and cross-functional decision-making.

Expected Solution

- **Industrial Data Ops:** Builds a structured data and metadata foundation to connect SOPs, P&IDs, datasheets, maintenance logs, and asset data.
- **AI Document Ingestion:** Converts scattered technical documents into a structured, searchable, and AI-ready knowledge base.
- **Q&A Assistant & Semantic Search:** Develops an AI assistant that helps engineers find trusted technical information through natural-language questions and context-based search.
- **Traceable Answer with Source:** Provides AI-generated answers with clear document sources, version references, approval status, and confidence indicators.
- **EDMS, AIMS, and Digital Twin Integration:** Connects to knowledge hub with EDMS, AIMS, and Digital Twin to provide technical information based on asset and operational context.
- **Failure Memory System and Recommendation:** Retrieves past failures, RCA, lessons learned, and recommended actions to support faster troubleshooting and more reliable execution.

Key Questions (Scope of Work)

Participants must address the following question in the deck, either within the proposed solution or business impact section.

1. **How can the Company build a structured Industrial Data Ops foundation to connect scattered plant knowledge sources?**
 - Main Focus: Build connected knowledge foundation.
2. **How can AI help engineers find trusted technical information faster and reduce the risk of improper execution?**
 - Main Focus: Improve search, Q&A, and answer verification.
3. **How can the Manufacturing Knowledge Hub be integrated with operational systems to support reliability, troubleshooting, and continuous improvement?**
 - Main Focus: Active knowledge for operational decision support.



Baseline Data

You can retrieve the baseline data from this link:

<https://bit.ly/CALIBER2026-TheCase>

Besides the baseline data, participants may propose additional knowledge of data sources beyond the provided baseline. However, every additional data source should be supported by a clear business justification. A well-justified, high-quality data architecture is preferred over simply integrating more data.

Data Source	Purpose
SOP & Technical Document (Datasheets)	Provides procedures and technical reference
P&IDs Data	Understand as-built equipment, piping systems, and instrumentation
Maintenance History	Capture incidents, root causes, and corrective actions
Tacit Knowledge	Capture SME experience and OPL data



CASE 2

Intelligence Manufacturing (Unified Dashboard, AI Insight, Root Cause Analysis, Follow-up Action Recommendation)

Topic

Manufacturing performance data is fragmented across multiple dashboards, systems, and reports, making it difficult to monitor plant performance, identify root causes, prioritize alerts, and track follow-up actions.

Background

In a chemical manufacturing plant, teams from Operations, Production, Energy, Maintenance, HSE, and Management use multiple dashboards and reports to monitor plant performance. However, each function often relies on different systems, KPI definitions, data formats, and reporting timelines. When an issue occurs, such as abnormal energy consumption, production loss, emission deviation, or equipment downtime, teams may spend significant time validating data, comparing reports, and aligning interpretations before deciding the next action.

This fragmented condition results in slower, more reactive decision-making. It also creates difficulties in identifying root causes, prioritizing critical alerts, recommending corrective actions, and tracking whether follow-up actions have been completed. As a result, potential optimization opportunities may be missed, and operational reliability can be affected.



Problem Statement

Participants are challenged to design an Intelligent Manufacturing Dashboard that integrates fragmented manufacturing data into a single pane of glass for decision-making.

The solution should connect data sources, display critical KPIs, identify operational problems, indicate possible root cause, prioritize alerts, and guide follow-up actions. The objective is to reduce time spent on data validation, accelerate problem-solving, improve operational efficiency, reduce downtime, and strengthen reliability and safety across functions.

Expected Solution

- **Rationalize & Integrate Dashboard:** Consolidates redundant dashboards and connects key manufacturing data sources into one governed view.
- **Single Pane of Glass:** Provides one integrated dashboard to monitor energy, production, emission, downtime, and operational performance.
- **Root Cause Analysis Support:** Helps identify probable causes of operational issues using problem history, KPI deviations, and data patterns.
- **Follow-up Action Recommendation:** Converts alerts and insights into prioritized actions with owner, guidance, and progress tracking.



Key Questions (Scope of Work)

Participants must address the following questions in the deck, either within the proposed solution or business impact section.

- 1. How can the Company rationalize fragmented dashboards and map manufacturing data sources into one governed data foundation?**
 - Main Focus: Reduce dashboard redundancy and create consistent KPI/data mapping across functions.
 - Key Capabilities: Data Source Mapping.

- 2. How can the Company build a single pane of glass that provides executive visibility of manufacturing performance?**
 - Main Focus: Provide one integrated dashboard to monitor operational performance.
 - Key Capabilities: Executive Level Dashboard.
 - Use Cases: Energy Forecasting, Similar-incident Retrieval, AI-based Root Cause Indication.

- 3. How can AI help identify probable root causes behind operational problems and translate insights into prioritized follow-up actions with clear ownership and tracking?**
 - Main Focus: Support faster problem-solving by connecting issue detection, prioritized alerts, recommended actions.
 - Key Capabilities: Problem Tank, Root Cause Indication, Alert Prioritization Engine, Action Tracking, and Action Guidance.

Baseline Data

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Besides the baseline data, participants may propose additional knowledge of data sources beyond the provided baseline. However, every additional data source should be supported by a clear business justification. A well-justified, high-quality data architecture is preferred over simply integrating more data.

Data Source	Purpose
Production Data	Monitor production output and performance
Incident Database	Trace and analyze past incidents
Equipment Performance	Monitor equipment condition and performance trends
Downtime Data	Track downtime causes and duration



FAQ (Case Book)

1. **Do we need to work on both use cases?**
No, please choose only one case to solve.
2. **Can we combine ideas from both cases?**
No. Select just one of the two cases; while extended solutions to the problem may be developed for whichever case you pick, your attention should remain on that particular case.
3. **Are the "Key Questions" listed under each case mandatory to answer individually?**
Yes, you are required to answer all the key questions in the proposed solution or business impact section.
4. **Will we be provided with real datasets, or should we use assumed data?**
All datasets required for the Preliminary Stage are included in this Case Book. You can use or generate dummy data that are relevant to your solution or with the provided all dataset. You may also supplement your analysis with publicly available data or reasonable assumptions where necessary.
5. **Can we use external or publicly available datasets in addition to the baseline data provided?**
Yes. You may use additional external or publicly available data sources beyond the baseline data provided, provided that the relevance and business value of each data source are clearly justified in the presentation deck.
6. **What file formats and size limits apply to the prototype, video, and presentation deck?**
For the Preliminary Stage, all 3 (three) materials combined should be no bigger than 10 MB. Submission format for Final Stage will be informed during the event.
7. **What language should be used for the presentation deck and pitch?**
All materials should be written and delivered in English.
8. **Can we use generative AI tools as part of our solution development process?**
The use of AI tools is permitted, provided they are used lawfully and in accordance with their applicable terms and conditions and privacy notices. All AI-generated outputs must be validated, iterated, and evaluated to ensure the quality, accuracy, and reliability of the resulting technology solutions before use or implementation.
9. **Can we update or replace our submission after it has been submitted?**
No. Only the first submission submitted by your team will be accepted and evaluated.
10. **What happens if we fail to submit one of the required deliverables?**
Your team will be disqualified and cannot continue to the next stage.

Chandra Asri Group CALIBER 2026 Committee

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